

Combinational Circuits-3

CS1400

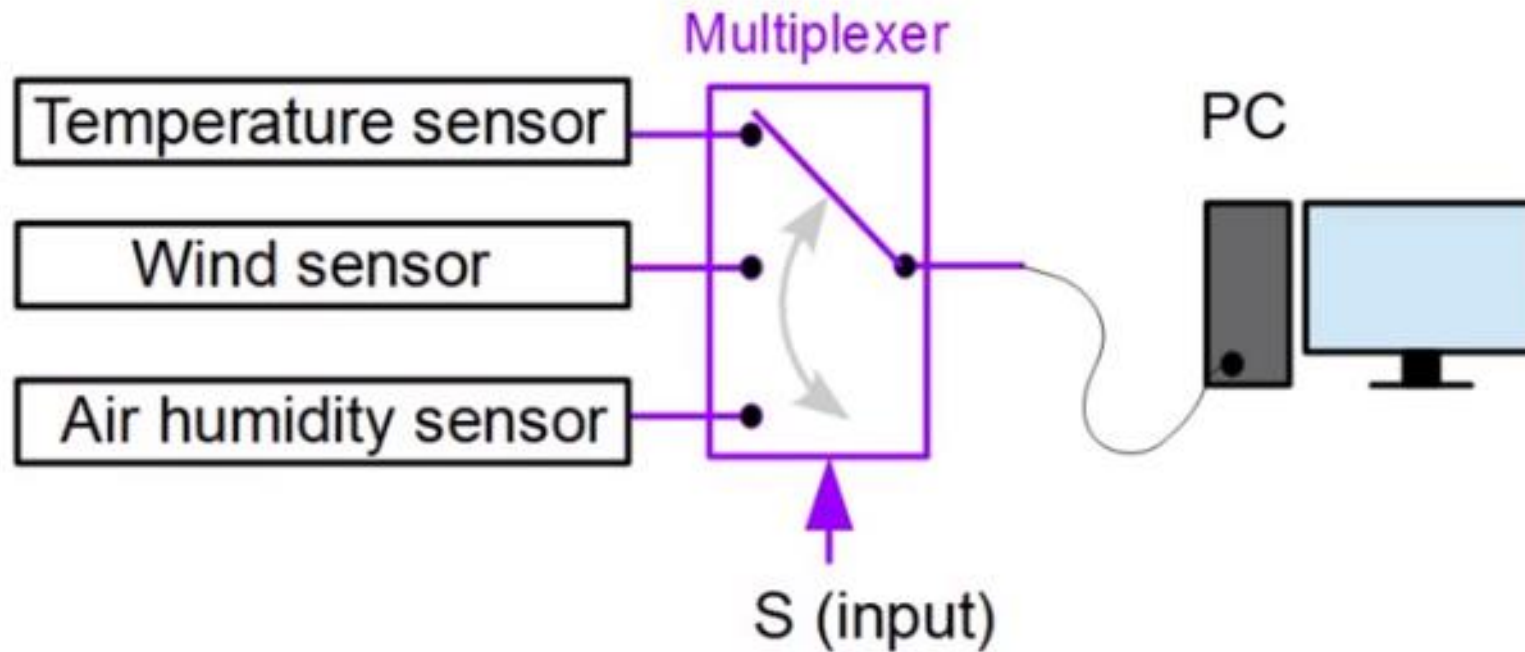
Spring 2022



Selection

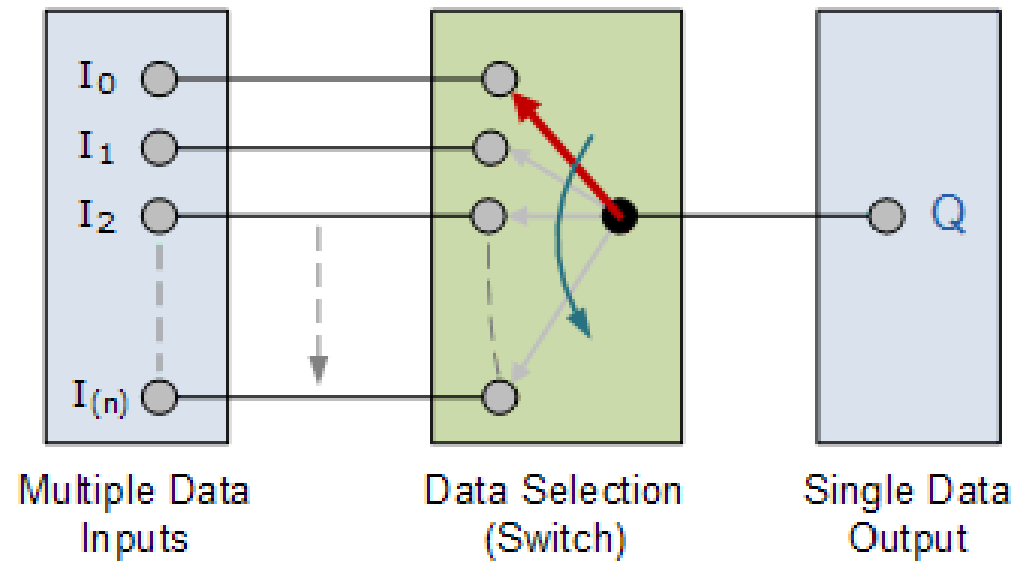


Selection



Multiplexer

- Multiplexer (data selectors) choose an output from among several possible inputs based on the value of a select signal.



2-input Multiplexer Design

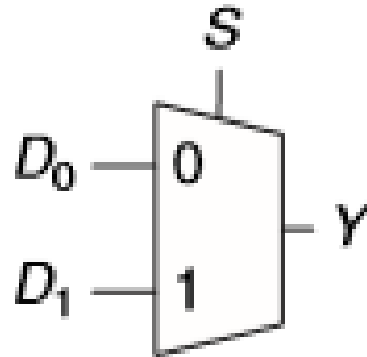
S	D1	D0	Y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1



2-input Multiplexer Design

S	D1	D0	Y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

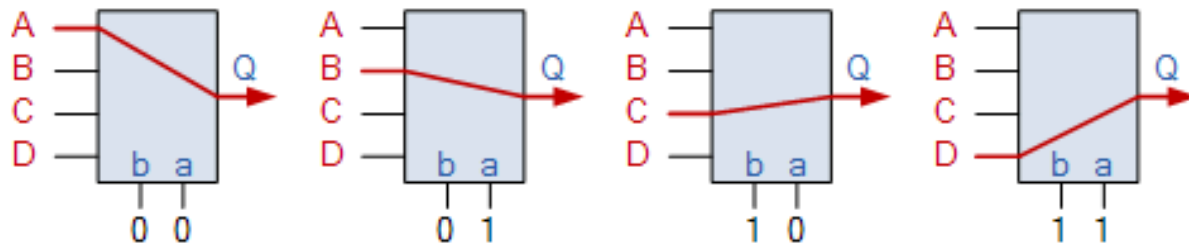
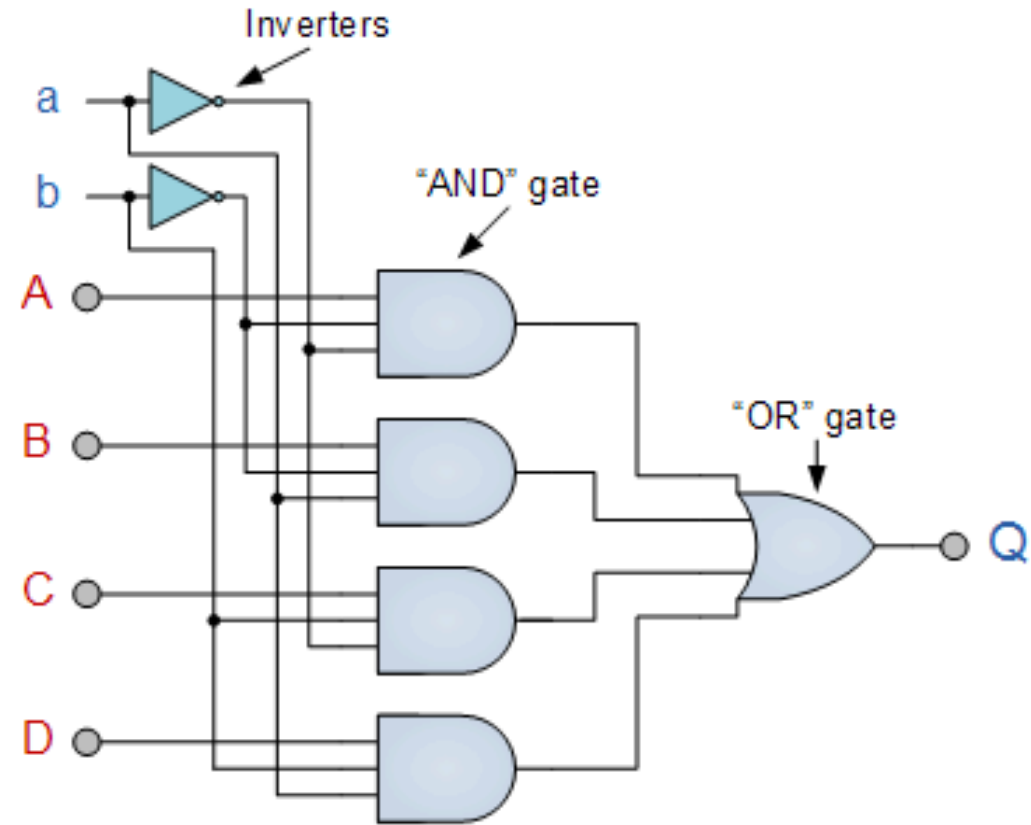
S - Selection bit
D1 - Input 1
D0 - Input 0
Y - Output



4 X 1 MUX

Truth Table

Select		Inputs				Q
b	a	D	C	B	A	
0	0	x	x	x	1	1
0	1	x	x	1	x	1
1	0	x	1	x	x	1
1	1	1	x	x	x	1



MUX

- Maximum of 2^n data inputs, 'n' selection lines and single output line.
- 4 X 1
- 8 X 1
- 16 X 1



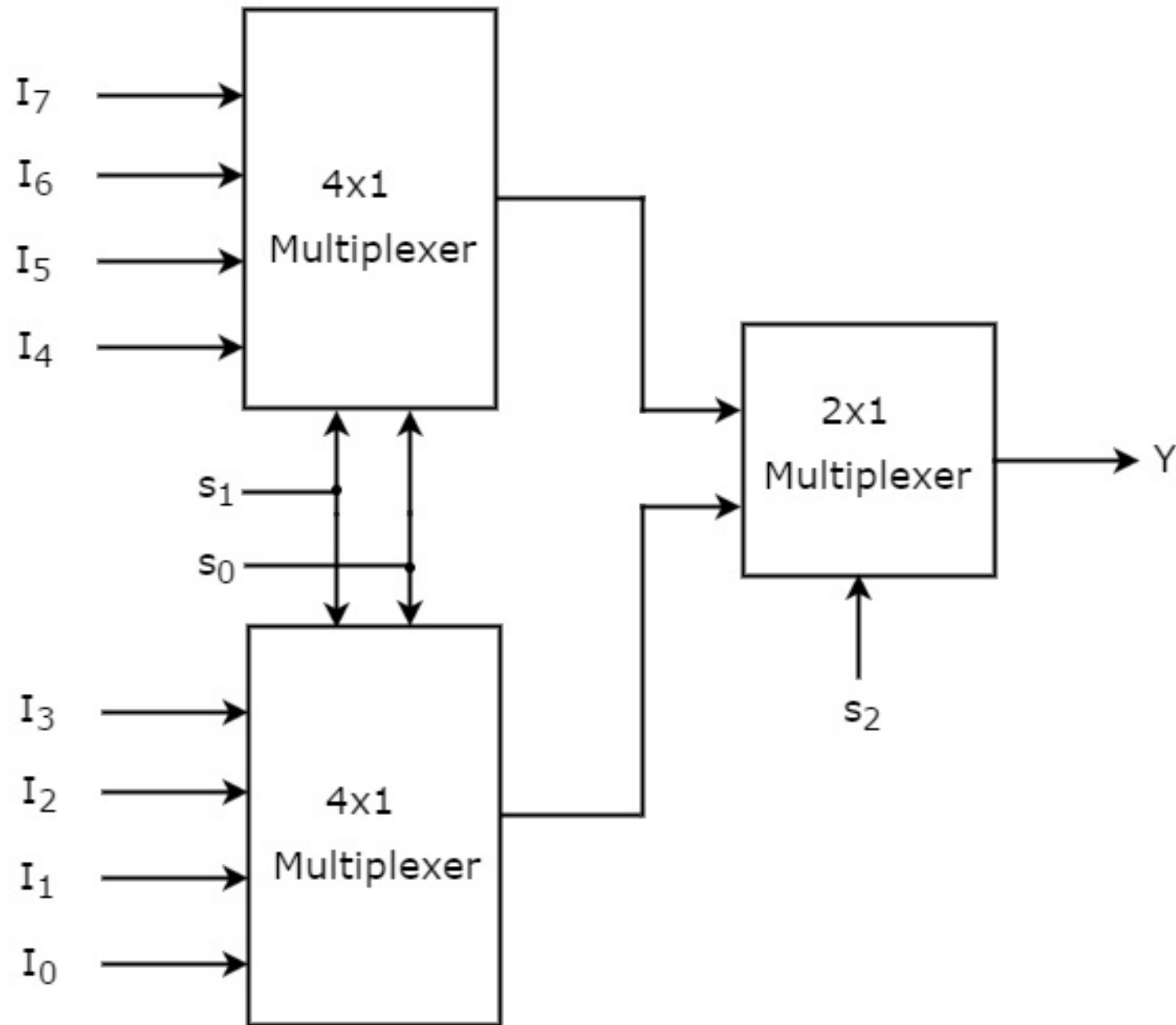
MUX

- Let the 8x1 Multiplexer has eight data inputs I_7 to I_0 , three selection lines s_2 , s_1 & s_0 and one output Y .

Selection Inputs			Output
s_2	s_1	s_0	Y
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

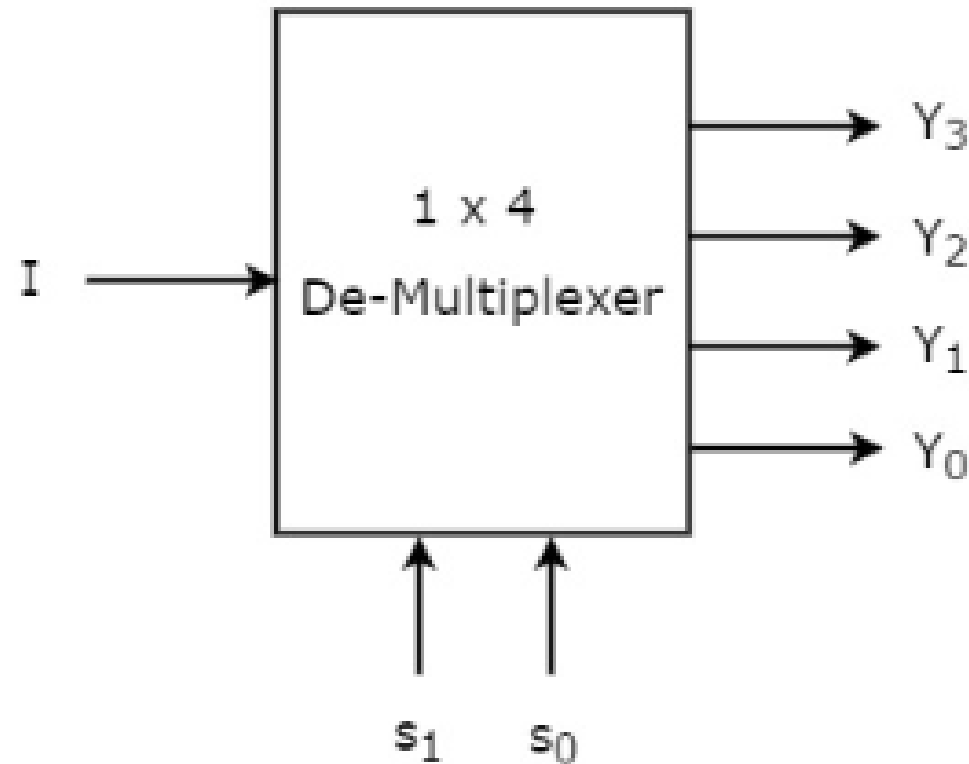


8 X 1 MUX



De-Multiplexer

- **De-Multiplexer** is a combinational circuit that performs the reverse operation of Multiplexer. It has single input, 'n' selection lines and maximum of 2^n outputs.



Decoder

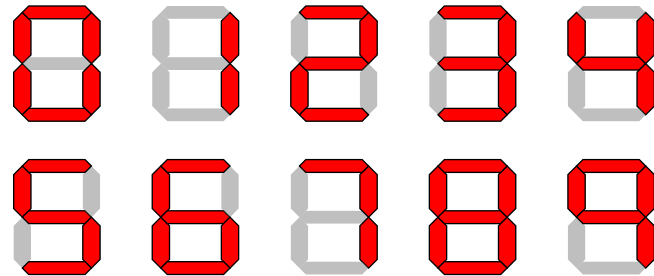
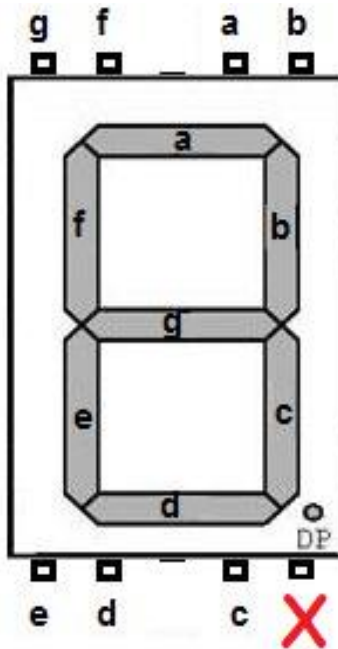
- A decoder is a combinational logic circuit which accepts inputs as binary numbers and activates its output in a given format.
- Decoder has 'n' inputs and can have maximum of 2^n unique outputs.
- They are used in a wide variety of applications, including data multiplexing and data demultiplexing, seven segment displays, and as address decoders for memory and port-mapped I/O.
- We will be learning how a seven segment decoder works.



DECODER

- **Decoder** is a combinational circuit that has 'n' input lines and maximum of 2^n output lines.

- Ex.

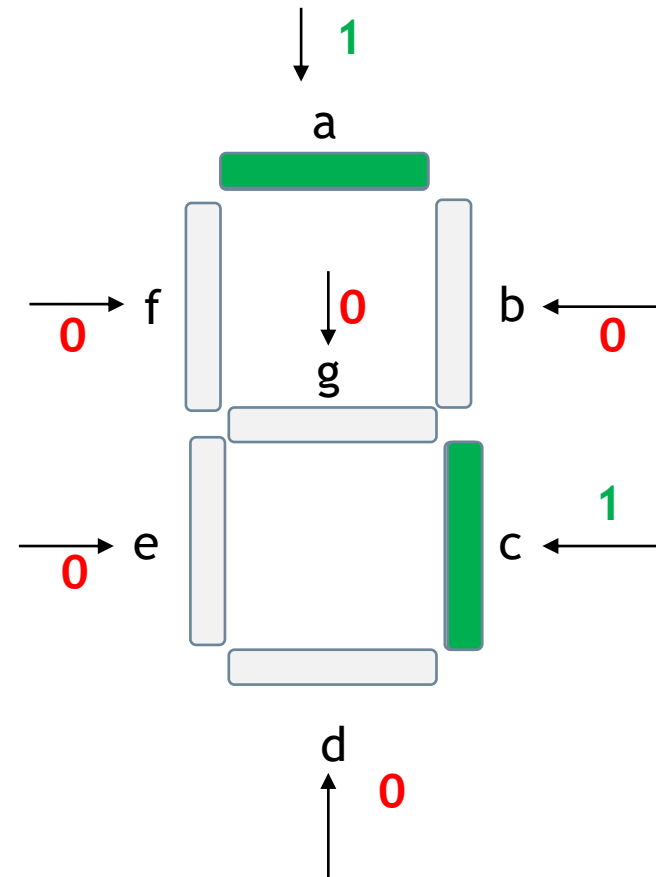


Binary Input	Decimal Display
0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7
1000	8
1001	9

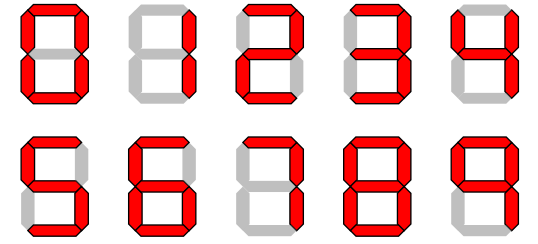


7 segment Display

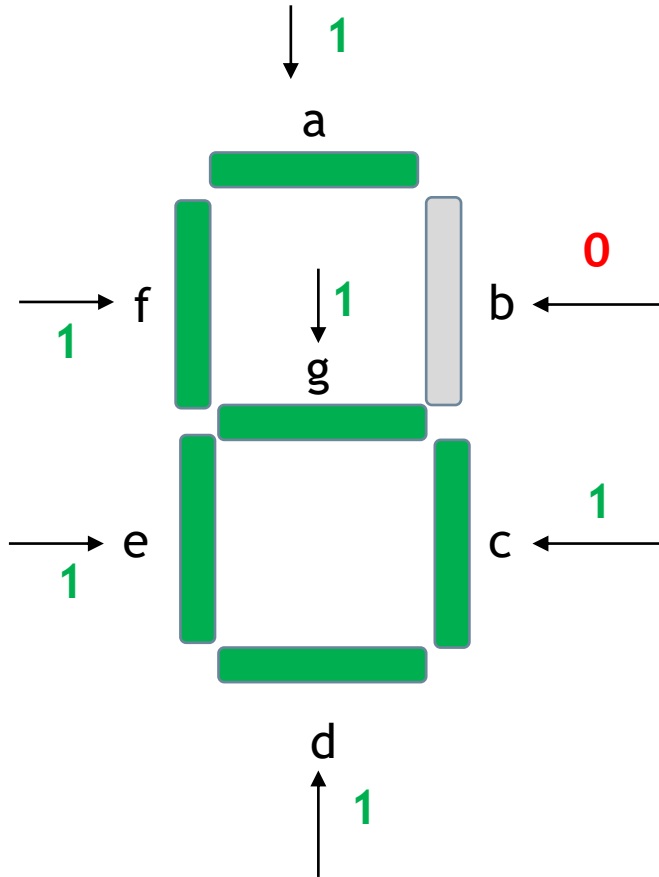
- When input is **1** The segment will be **ON** .
- When input is **0** the segment will be **OFF**.
- Example **a=1,b=0,c=1,d=0,e=0,f=0**



7 segment Display



- What will be the truth table for displaying 6 is seven segment display.

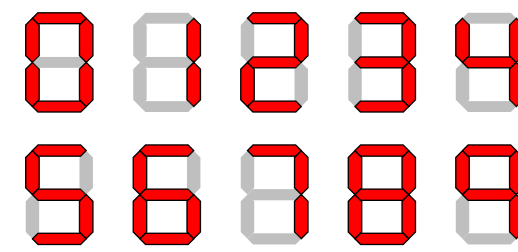


a	b	c	d	e	f	g
1	0	1	1	1	1	1

Write the truth table to display 0 to 9



Displaying all numbers 0 -9



- We need to display 10 numbers 0 to 9, so we take binary equivalent of 0 to 9 as inputs.

INPUTS					OUTPUTS						
	A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0							
3	0	0	1	1							
4	0	1	0	0							
5	0	1	0	1							
6	0	1	1	0							
7	0	1	1	1							
8	1	0	0	0							
9	1	0	0	1							



For Segment 'a'

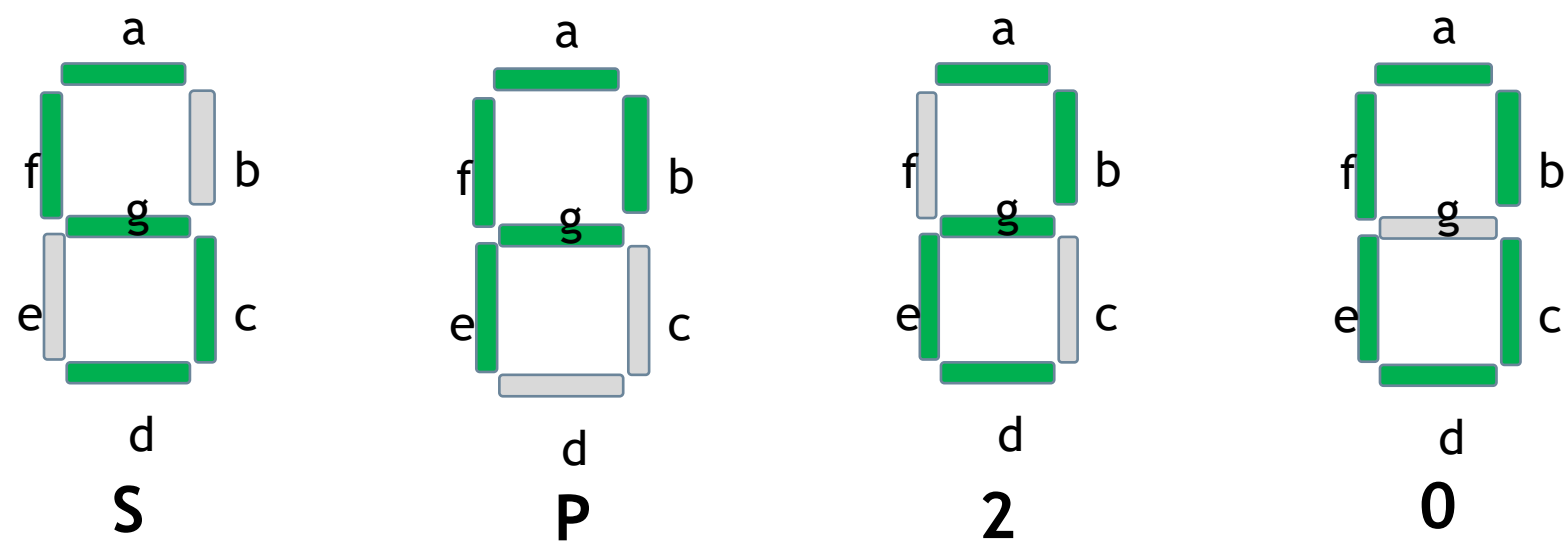
A	B	C	D	a
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1

CD\AB	A'B'	A'B	AB	AB'
C'D'	1			1
C'D		1		1
C D	1	1		
C D'	1	1		

$$A'C + AB'C' + A'BD + B'C'D'$$



Displaying all numbers SP20



ASCII - Binary Character Table
will give the binary
equivalent of letters

	INPUTS								OUTPUTS							
	I1	I2	I3	I4	I5	I6	I7	I8								
S	0	1	0	1	0	0	1	1								
P	0	1	0	1	0	0	0	0								
2	0	0	0	0	0	0	1	0								
0	0	0	0	0	0	0	0	0								

For this we need an 8 variable K -Map.(Big one)
Here in this course we are only considering the maximum of 4 variable K-Map ,



Design Circuit

- For **demo purpose** I am taking the inputs as two variable combinations to display four numbers. By taking 2 variables we get 4 different input combinations.
- **Let us assume** 0 is represented by binary 0 0, S is represented by binary 0 1, 2 is represented by binary 1 0, P is represented by binary 1 1.

	A	B
0	0	0
S	0	1
2	1	0
P	1	1

a	b	c	d	e	f	g
1	1	1	1	1	1	0
1	0	1	1	0	1	1
1	1	0	1	1	0	1
1	1	0	0	1	1	1



Circuit for segment 'a'

A	B	a
0	0	1
0	1	1
1	0	1
1	1	1

B\A	A'	A
B'		
B		



Circuit for segment 'b'

A	B
0	0
0	1
1	0
1	1

b
1
0
1
1

B\A	A'	A
B'		
B		

Expression for segment b =



Circuit for segment 'c'

A	B	c
0	0	1
0	1	1
1	0	0
1	1	0

B\A	A'	A
B'		
B		

Expression for segment c =



Circuit for segment 'd'

A	B	d
0	0	1
0	1	1
1	0	1
1	1	0

B\A	A'	A
B'		
B		

Expression for segment d =



Circuit for segments e, f and g

- For 'e', 'f' and 'g'. Do it your self. I will give you 10 min.

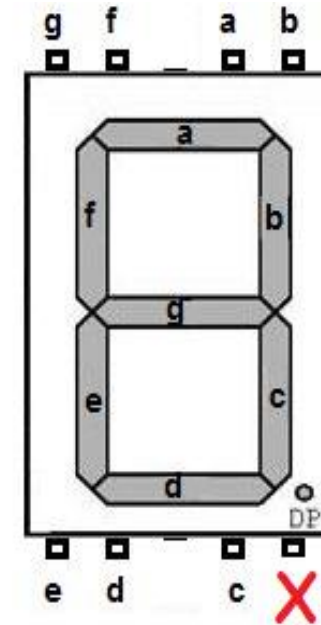
	A	B
0	0	0
S	0	1
2	1	0
P	1	1

a	b	c	d	e	f	g
1	1	1	1	1	1	0
1	0	1	1	0	1	1
1	1	0	1	1	0	1
1	1	0	0	1	1	1



Expressions for each segments.

- Segment 'a' =
- Segment 'b' =
- Segment 'c' =
- Segment 'd' =
- Segment 'e' =
- Segment 'f' =
- Segment 'g' =



Worksheet 07

- Complete worksheet 07





To Do as of 03/107

- Complete Worksheet 7
- Next class Sequential Circuits

